

Multivitamins in the Prevention of Cancer in Men: The Physicians' Health Study II Randomized Controlled Trial



Context Multivitamin preparations are the most common dietary supplement, taken by at least one-third of all US adults. Limited observational studies have not provided evidence regarding associations of multivitamin use with total and site-specific cancer incidence or mortality. **Objective** To determine whether long-term multivitamin supplementation decreases the risk of total and site-specific cancer events among men. **Design** The Physicians' Health Study II is a randomized, double-blind, placebo-controlled trial of a common multivitamin that began in 1997 with treatment and follow-up through June 1, 2011. **Setting and Participants** A total of 14,641 male U.S. physicians initially aged ≥ 50 years (mean $\pm$ SD age; 64.3 $\pm$ 9.2 years), including 1,312 men with a history of cancer at randomization, were enrolled. **Intervention** Daily multivitamin, as Centrum Silver. **Main Outcome Measures** A primary outcome was total cancer (excluding non-melanoma skin cancer), with prostate, colorectal, and other site-specific cancers among secondary endpoints included in this report. **Results** During a median (interquartile range) follow-up of 11.2 (10.7 to 13.3) years, there were 2,669 men with confirmed cancer, including 1,373 cases of prostate cancer and 210 cases of colorectal cancer. Compared with placebo, men taking a daily multivitamin had a statistically significant reduction in the incidence of total cancer (active and placebo multivitamin groups, 17.0 and 18.3 events, respectively, per 1,000 person-years; hazard ratio [HR], 0.92; 95% confidence interval [CI], 0.86–0.998; $P=0.044$). There was no significant effect of a daily multivitamin on prostate cancer (HR, 0.98; 95% CI, 0.88–1.09; $P=0.76$), colorectal cancer (HR, 0.89; 95% CI, 0.68–1.17; $P=0.39$), or other site-specific cancers. There was a lower risk of cancer mortality that did not reach statistical significance (HR, 0.88; 95% CI, 0.77–1.01; $P=0.07$). Daily multivitamin use was associated with a reduction in total cancer among 1,312 men with a baseline history of cancer (HR, 0.73; 95% CI, 0.56–0.96; $P=0.022$), but this did not differ significantly from that among 13,329 men initially free of cancer (HR, 0.94; 95% CI, 0.87–1.02; $P=0.15$) (P , interaction = 0.07). **Conclusions** In this large prevention trial of male physicians, daily multivitamin supplementation modestly but significantly reduced the risk of total cancer.